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...rable equation

Example 2.2

$$\frac{dy}{dt} = \frac{y\cos t}{1+2y^2} \qquad y(0)=1$$

Here $\frac{dy}{dt} = F(t)g(y)$ where $F = \cos t$ $\quad g = \frac{y}{1+2y^2}$

Special solution

$g(y)=0$ when $y=0$ ∴ we have a special solution, $\forall t$

Typical solution

Assuming $y \neq 0$ we rewrite the ODE as

$$\int \frac{(1+2y^2)}{y}\,dy = \int \cos t\,dt$$

$$\Rightarrow \int \frac{1}{y} + 2y\,dy = \sin t + c$$

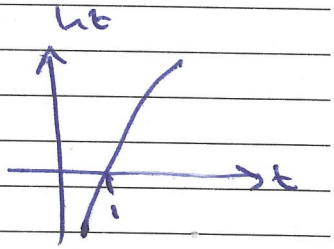
$$\Rightarrow \boxed{\ln|y| + y^2 = \sin t + c}$$ typical sol^n

Now we use the initial condition to find c.

$y(t=0)=1 \Rightarrow \ln|1| + 1^2 = \sin 0 + c$

$0 \qquad\qquad 0$

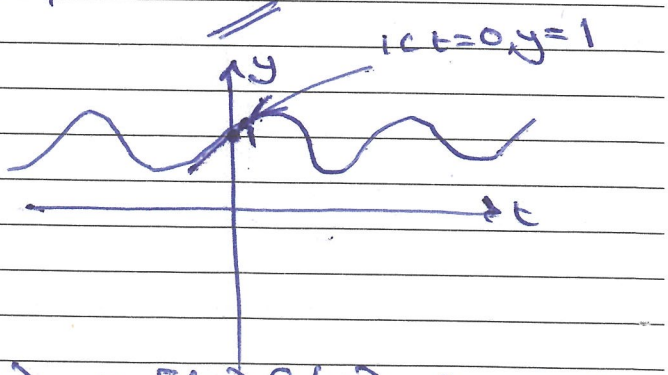
$\therefore c = 1$



The solution of the IVP is

$$\ln|y| + y^2 = \sin t + 1 \quad , \quad \forall t$$

- IC $t=0 \Rightarrow y=1$
- $\frac{dy}{dt} = \frac{y\cos t}{1+2y^2} > 0$
- curve doesn't cross $y=0$
- oscillates in t



Test

$\frac{dy}{dt} = f(t,y) = F(t)g(y)$ if separable

$$f(a,b)f(c,d) = F(a)g(b)F(c)g(d)$$
$$= F(a)g(d)F(c)g(b)$$
$$= f(a,d)f(c,b)$$

Example 2.4 $\quad t\frac{dy}{dt} = t + y$

For $t \neq 0$ $\quad \frac{dy}{dt} = 1 + \frac{y}{t} = 1 + v$ where $v = y/t$

Then $y = vt \Rightarrow \frac{dy}{dt} = v + t\frac{dv}{dt}$

Substitute:

$$v + t\frac{dv}{dt} = 1 + v$$

$$\int dv = \int \frac{dt}{t}$$

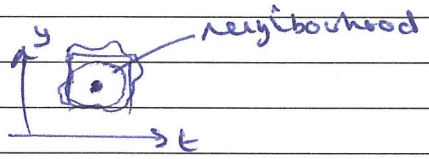
$$\Rightarrow v = \ln|t| + c$$

$$\Rightarrow y = t\ln|t| + ct \quad \text{since } y = vt$$

valid for $t < 0$ or $t > 0$

Existence & Uniqueness





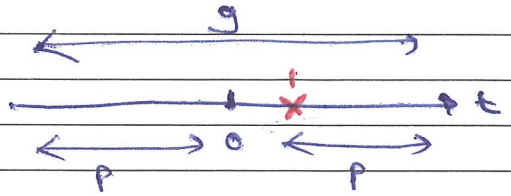
Example 2.10

$$t\frac{dy}{dt} + 2y = 4t^2$$

$$\frac{dy}{dt} + \frac{2y}{t} = 4t \qquad \text{linear first order ODE}$$

$p(t) = \frac{2}{t}$ $\quad g(t) = 4t$

continuous for $t < 0$ or $t > 0$ $\quad$ continuous everywhere $(\forall t)$



The initial condition $y(t_0 = 1) = 2$ has $t_0 = 1 > 0$ and so by theorem 2.6 there is a unique solution to the IVP on the interval $0 < t < \infty$

Integrating factor $\quad \mu = e^{\int p\,dt} = e^{\int \frac{2}{t}dt} = e^{2\ln|t|} = e^{\ln t^2} = t^2$

$$\frac{d}{dt}(yt^2) = 4t^3$$

$$\Rightarrow yt^2 = \int 4t^3\,dt = t^4 + c$$

$$\Rightarrow y = t^2 + c/t^2 \text{ is the general solution, } t \neq 0.$$

IVP $y(t_0 = 1) = 2 \Rightarrow \quad 2 = 1^2 + C/1 \quad \Rightarrow C = 1$

$\therefore \; y = t^2 + 1/t^2$ for $t > 0$

